

Exercices Lab 2 Bonus

LCG

PRNGs (6/6)



- Homework Exercises

✓ You are given the following LCG parameters:

- $x_0 = 51357953340700960539503091666792115306$
- $x_1 = 60122000323363606827929954412280678934$
- $x_2 = 45876883472656003592807994096015528525$
- $m = 2^{127} - 1$

Recover a and b ☺

✓ Let the following LCG parameters:

- $x_0 = 14$
- $a = 193$
- $b = 78$
- $m = 1337$

Compute the **period** of the LCG. In other words, how many distinct values does it generate?

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1 LCG Parameter Recovery

1.1 Problem Statement

Given an LCG (Linear Congruential Generator) with recurrence relation:

$$x_{n+1} = (a \cdot x_n + b) \mod m$$

where:

$$m = 2^{127} - 1$$

$$x_0 = 51357953340700960539503091666792115306$$

$$x_1 = 60122000323363606827929954412280678934$$

$$x_2 = 45876883472656003592807994096015528525$$

Recover the parameters a and b .

1.2 Solution Method

1.2.1 Step 1: Set up equations

From the recurrence relation:

$$x_1 \equiv a \cdot x_0 + b \pmod{m} \quad (1)$$

$$x_2 \equiv a \cdot x_1 + b \pmod{m} \quad (2)$$

1.2.2 Step 2: Eliminate b

Subtract equation (1) from equation (2):

$$x_2 - x_1 \equiv a \cdot (x_1 - x_0) \pmod{m}$$

Let:

$$d_1 = x_1 - x_0$$

$$d_2 = x_2 - x_1$$

Then:

$$a \equiv d_2 \cdot d_1^{-1} \pmod{m}$$

1.2.3 Step 3: Compute modular inverse

Since $m = 2^{127} - 1$ is prime, we use Fermat's Little Theorem:

$$d_1^{-1} \equiv d_1^{m-2} \pmod{m}$$

1.2.4 Step 4: Compute a

$$a = d_2 \cdot d_1^{-1} \pmod{m} = 123279044536326290368097455220904336127$$

1.2.5 Step 5: Compute b

From equation (1):

$$b = x_1 - a \cdot x_0 \pmod{m} = 35943085179036080968431775984807083513$$

1.3 Python Implementation

```
1 def solve_lcg():
2     # Given values
3     m = 2**127 - 1
4     x0 = 51357953340700960539503091666792115306
5     x1 = 60122000323363606827929954412280678934
6     x2 = 45876883472656003592807994096015528525
7
8     # Compute differences
9     d1 = (x1 - x0) % m
10    d2 = (x2 - x1) % m
11
12    # Compute modular inverse of d1 using Fermat's Little Theorem
13    # Since m is prime: d1^(-1) ≡ d1^(m-2) mod m
14    inv_d1 = pow(d1, m-2, m)
15
16    # Compute a
17    a = (d2 * inv_d1) % m
18
19    # Compute b
20    b = (x1 - a * x0) % m
21    return a, b
```

1.4 Final Solution

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SOLUTION:

a = 123279044536326290368097455220904336127

b = 35943085179036080968431775984807083513

$$a = 123279044536326290368097455220904336127$$

$$b = 35943085179036080968431775984807083513$$

2 LCG Period Analysis

2.1 Problem

Given LCG parameters:

$$\begin{aligned}x_0 &= 14 \\ a &= 193 \\ b &= 78 \\ m &= 1337\end{aligned}$$

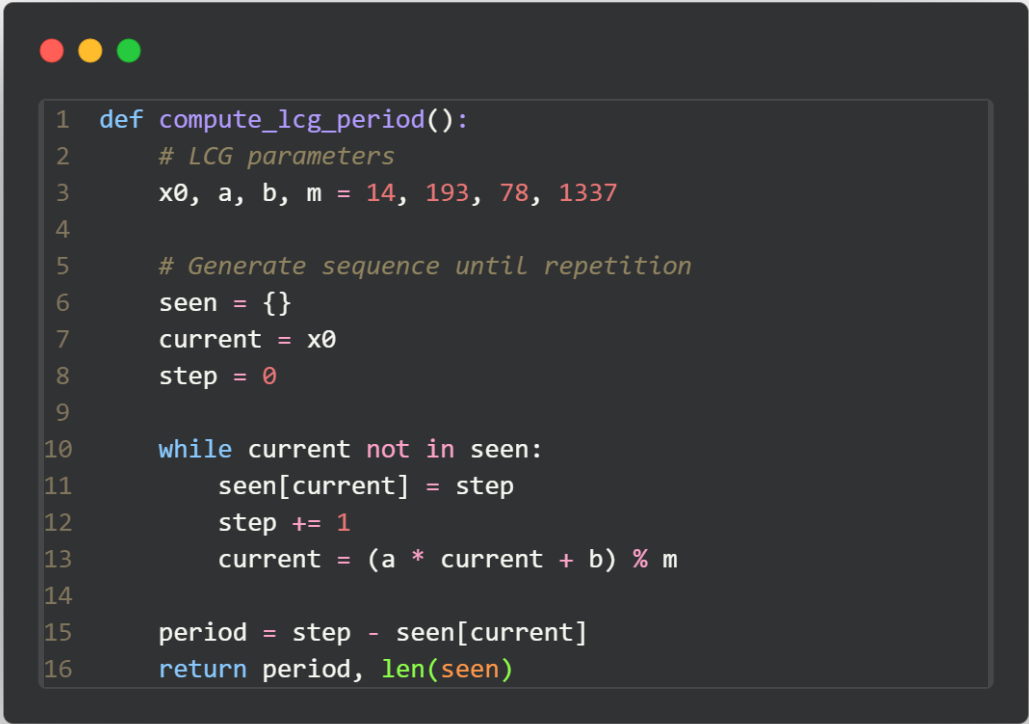
Compute the period (number of distinct values generated).

2.2 Solution

The LCG recurrence:

$$x_{n+1} = (193 \cdot x_n + 78) \mod 1337$$

2.2.1 Python Implementation



```
1 def compute_lcg_period():
2     # LCG parameters
3     x0, a, b, m = 14, 193, 78, 1337
4
5     # Generate sequence until repetition
6     seen = {}
7     current = x0
8     step = 0
9
10    while current not in seen:
11        seen[current] = step
12        step += 1
13        current = (a * current + b) % m
14
15    period = step - seen[current]
16    return period, len(seen)
```

2.3 Answer

```
Period: 285  
Distinct values: 285
```

285

The LCG has period 285, generating 285 distinct values before repeating.